Baby rat pathing

Greedy · Visible BFS · DistBug

How a 1×1 baby rat takes one step toward cheese or the published king. North is up. The rat sees a 90-degree cone with radius squared 20.

On each movement turn

Visible BFS	Empty-handed, walking toward a pile the rat can already see.
DistBug	Carrying enough cheese, walking toward the published king.
Greedy	Fallback inside both helpers, and the 8-neighbor probe they share.
Wander	No published king, or no pile to chase.

Map key and where each helper runs

Diagrams match the team tests: north up, +x east.

R	G	#	?	O	.
Rat	Goal	Wall or dirt	Not seen this turn	Other robot	Open tile

? is a tile the rat has not seen. O is another robot on a passable tile, not a wall.

Cheese chase → Visible BFS

- The pile is chosen from cheese the rat currently sees.
- BFS can walk around a wall in that cone.
- Tiles outside the cone are not on the path.
- If there is no seen path, it uses greedy.

King return → DistBug

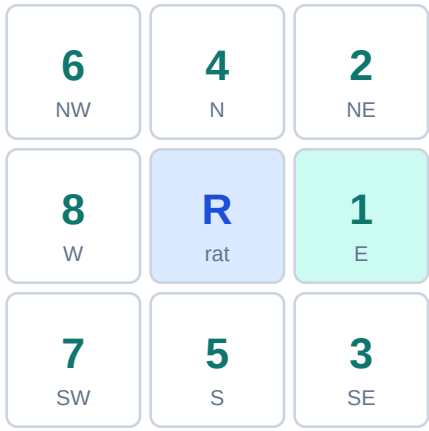
- The king publishes its center. It is often far outside the 90-degree cone.
- BFS cannot invent a path through unseen tiles.
- DistBug hugs a wall it has seen and keeps hugging next turn.
- Until it sees a wall, it walks greedy.

Greedy — one step

Used as the fallback. Does not look at map info, only which neighbors are legal.

1. If the rat is already on the goal, do nothing.
2. Take the compass direction toward the goal.
3. Build an 8-step probe: straight, then left/right pairs, then the opposite.
4. Take the first legal neighbor.
5. If every neighbor is illegal, turn to face the goal (if turning is allowed).

Goal east: probe order



First legal number wins. Goal is east, so 1 is straight at G.

Greedy along a wall it has not planned around

Goal is north. North, northwest, and northeast are blocked. West and east are open.

Wall north of the rat. Greedy tries N, then NW, then NE, then west before east.

Step: west, not east.

Why west?

- Straight at the goal is north — blocked.
- Left of north is northwest — blocked.
- Right of north is northeast — blocked.
- Next left is west — open. That is the move.
- East is also open, but east is later in the probe.
- If the goal is only just over the wall, the next turn can walk back east. That is the ping-pong.

Visible BFS — one step

Only tiles the rat saw this turn are walkable. Unknown tiles are not a path.

1. If the rat is already on the goal, do nothing.
2. Collect this turn's passable seen tiles, plus the tile the rat is standing on.
3. If almost nothing is seen, use greedy.
4. Target is the goal if that tile was seen and passable. Otherwise the seen passable tile closest to the goal.
5. Breadth-first search on those seen tiles only (8 neighbors, if both ends were seen).
6. Walk the first step of that path, if that step is legal.
7. If there is no seen path, use greedy.

Seen tiles only

BFS may walk the southern open tiles. The northern ? tiles are not in the search.

BFS walks a corridor it can see

Greedy will take an unseen shortcut if that neighbor looks legal.

Greedy

East is the wall. Next greedy probe is northeast — the unseen shortcut.

Likely first step: northeast into ?

Visible BFS

Northern tiles were never seen, so they are not on the BFS path.

First step: southeast into the seen corridor

DistBug — one step

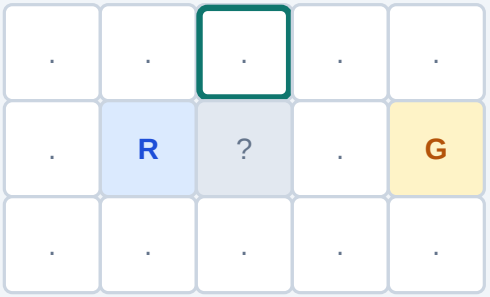
Remembers a wall across turns. Leaves only when closer than the first hit.

1. If already on the goal, stop following and do nothing.
2. If the goal changed, drop the old wall-follow and start over.
3. If currently following: if the way toward the goal is open and the rat is closer than when it first hit the wall, walk that way and stop following.
4. If still following, hug the wall (sweep toward the chosen hand; prefer a step that matches current facing).
5. If not following: if the tile toward the goal is blocked, and the rat has seen a wall, dirt, or map edge there, start following and hug.
6. Otherwise walk greedy. A blocked tile the rat has not seen does not start a follow. Another robot on a passable tile is not a wall.

DistBug only hugs a wall it has seen

Left-hand follow if left is open or right is not. Facing into the wall makes the first open left-hand step northeast.

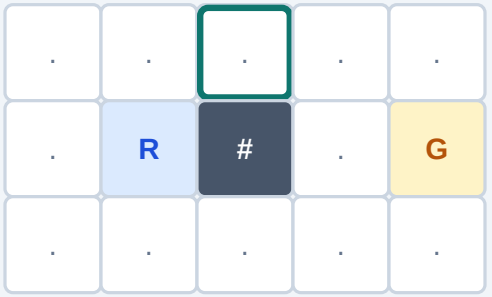
Blocked, but not seen



East is blocked. DistBug does not start follow. Greedy may step northeast.

No follow next turn.

Seen wall on the way to the goal



Rat faces east. DistBug starts follow and steps northeast along the wall.

Still following on the next turn.

Leaving the wall, and what is not a wall

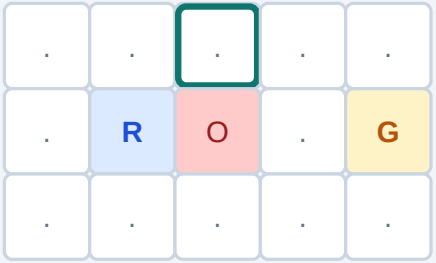
DistBug stops only when closer than the recorded hit and the way to the goal is open.

Leave

Already following. First hit was farther away. East is open and the rat is closer, so walk toward the goal and stop hugging.

Closer-than-hit is what stops walking around a U-shaped wall forever.

Another robot is not a wall



East occupied. DistBug does not hug. Next open greedy step is northeast.

What to remember

All three take one step per turn. Standing on the goal never moves or turns.

Helper	Idea	Used for
Greedy	First legal neighbor toward the goal.	Open ground; fallback.
Visible BFS	Shortest path on tiles seen this turn.	Cheese pile already in the cone.
DistBug	Greedy until a seen wall, then hug until closer.	King often outside the cone.

- BFS is not greedy. It searches seen tiles. Greedy only ranks the eight neighbors.
- BFS will not walk a tile the rat has not seen, even if that step looks open.
- DistBug will not hug a wall it has not seen, and will not hug another rat.
- Cheese → Visible BFS. King → DistBug. Both call greedy when they have nothing better.